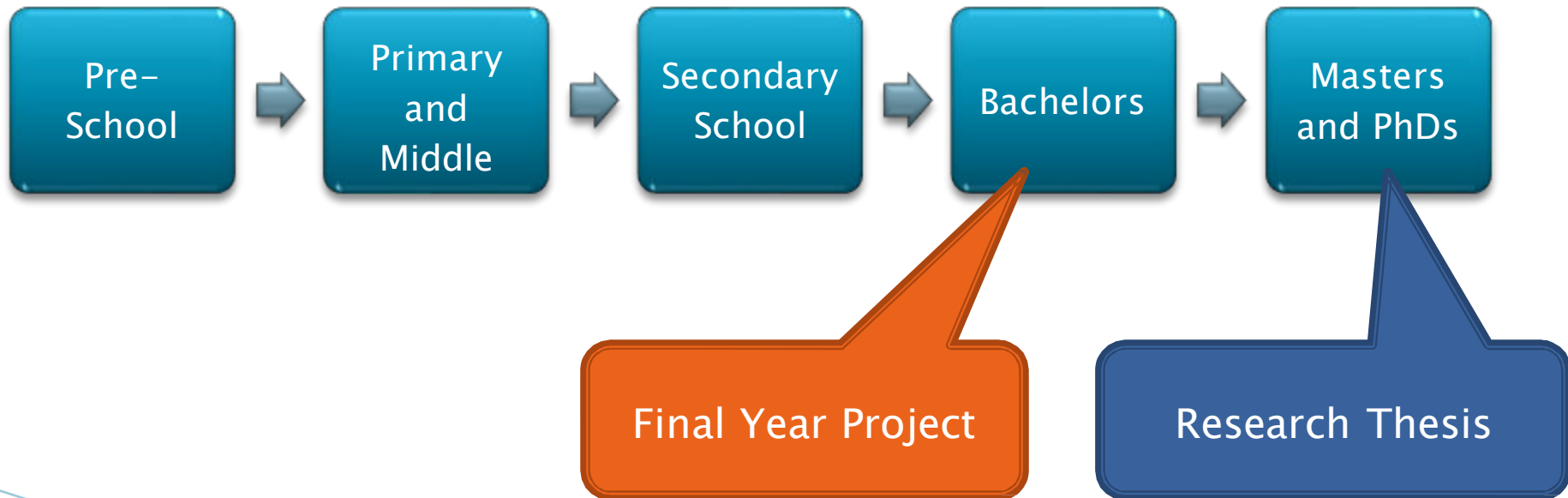


Week 01: Topic-01

Introduction to Research

Research Methodologies

Background



Definition of Research

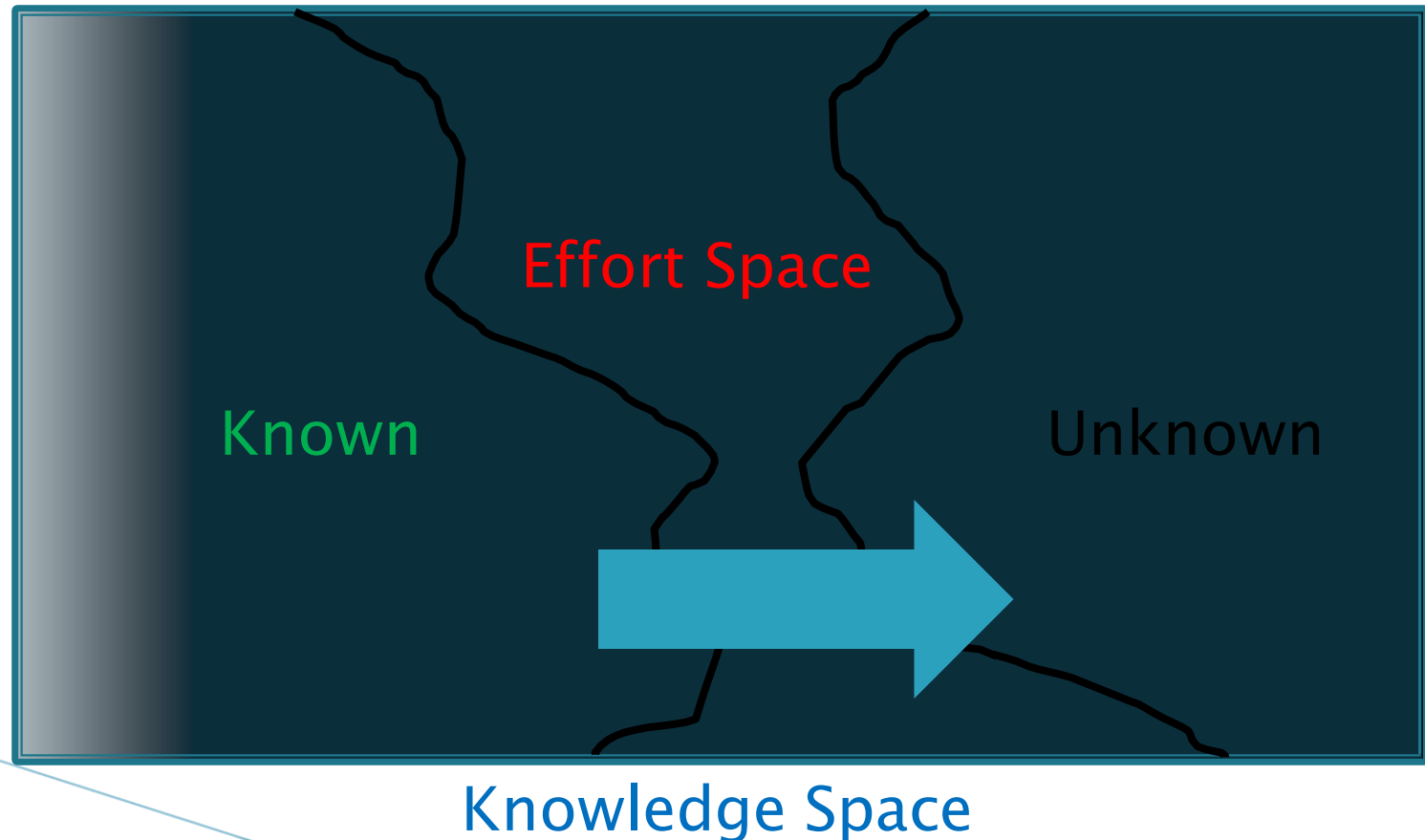
- ▶ Research in common words refers to a search for knowledge. Re + search
- ▶ Definition 1 : “Research is a scientific and systematic search for pertinent information on a specific topic”
- ▶ Definition 2 : “Research is a careful investigation or inquiry specially through search for new facts in any branch of knowledge” ¹
- ▶ Definition 3 : “Research is a systematized effort to gain new knowledge” ²

¹The Advanced Learner’s Dictionary of Current English

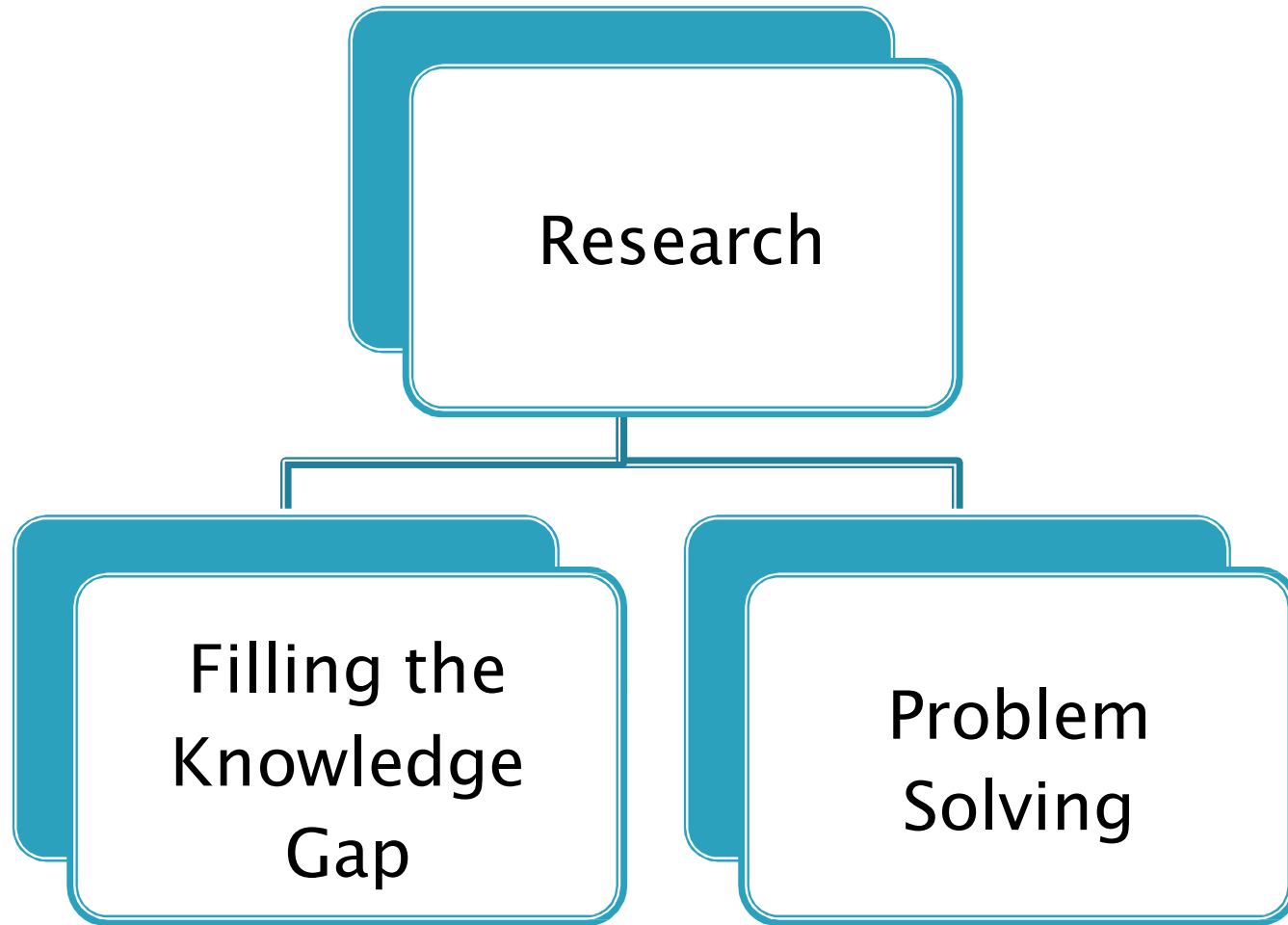
²Redman and Mory

Research – voyage of discovery

- ▶ a movement from the known to the unknown.



Understanding Research



Filling the Knowledge Gap – Example

► Amicable Numbers

- Amicable numbers are two different numbers related in such a way that the sum of the proper divisors of each is equal to the other number.
- The smallest pair of amicable numbers is
 - (220 and 284).

Proper divisors of 220 are 1, 2, 4, 5, 10, 11, 20, 22, 44, 55 and 110, of which the sum is 284

Proper divisors of 284 are 1, 2, 4, 71 and 142, of which the sum is 220.



Thabit ibne Qurra

In 9th Century, found two more pairs...



Pierre de Fermat

In 1636,
(17,296 and 18,416)
In 1638,
(9,363,584 and 9,437,056)



René Descartes

In 1638,



Leonhard Euler

In 1750,
found 58 more pairs

- By 1946, there were 390 known pairs, before the advent of computers
- As of January 2020, there are over 1,225,063,681 known amicable pairs.

Objectives Of Research

Research Methodologies

Objectives of Research

- ▶ The purpose of research is to discover answers through the application of scientific procedures.
- ▶ Though each research study has its own specific purpose, we may think of research objectives as falling into a number of following broad groupings
 - To gain familiarity with a phenomenon or to achieve new insights into it **Exploratory or Formulative Research.**
 - To portray accurately the characteristics of a particular individual, situation or a group **Descriptive Research.**
 - To determine the frequency with which something occurs or with which it is associated with something else **Diagnostic Research.**
 - To test a hypothesis of a causal relationship between variables – **Hypothesis–Testing or Causal Research.**

Exploratory research

- ▶ Exploratory research is “the preliminary research to clarify the exact nature of the problem to be solved.”
- ▶ The objective of exploratory research is to gather preliminary information that will help formulate problems precisely and develop hypotheses.
- ▶ Exploratory research is used when the topic or issue is new and when data is difficult to collect.
- ▶ It is often referred to as grounded theory approach or interpretive research as it used to answer questions like what, why and how.
- ▶ Example
 - Will increasing the variety of programs results in increase in number of students?

Descriptive research

- ▶ The objective of descriptive research is to describe the characteristics of various aspects of the problem under study
- ▶ For example, we are interested in studying student behavior i.e. listening to music while studying. We would be interested to know that
 - How popular this behavior really is?
 - What type of music they usually listen to?
 - Do they listen to music while studying certain subjects but not others?
- ▶ We will collect data through surveys, interviews etc.
- ▶ We can describe about this behavior in many different ways but we cannot anything conclusive about it.
 - Whether or not this is a good/bad behavior?
- ▶ Descriptive studies is usually followed by other types detailed/focused research studies

Diagnostic Research

- ▶ To determine the frequency with which something occurs or with which it is associated with something else
- ▶ Commonly used in medical research
 - E.g. Baby with rashes on her arms and her mother said she recently introduced cow milk in her diet.
 - The doctor is faced with a challenge to determine whether this baby has developed cow milk allergy or not?
 - Doctor need to conduct diagnostic research
 - Study various articles
 - Conduct different tests
 - Consult with dietitian

Causal research

- ▶ The objective of causal research is to test hypotheses about cause-and-effect relationships.
 - i.e. to determine which variable might be causing a certain behavior
- ▶ It is also known as explanatory research.
- ▶ What variations take place in an dependent variable with the change in the independent variable.
- ▶ Correlation among two variables does not imply causation...
 - Need to be careful with hidden variables
- ▶ For example, we are interested to know that whether or not listening to music have any effect on students performance?
- ▶ Experiment
 - Divide student into two groups and assign them some task e.g. memorization
 - Group 1...doing this task while listening to music
 - Group 2...doing this task without listening to music

Motivations and Essential Skills for Research

Research Methodologies

Research Motivations

- ▶ What makes people to undertake research?
- ▶ Possible motives for doing research may be either one or more of the following:
 - Desire to face the challenge in solving the unsolved problems, i.e., concern over practical problems initiates research;
 - Desire to get a research degree along with its consequential benefits;
 - Desire to get intellectual joy of doing some creative work;
 - Desire to be of service to society;
 - Desire to get respectability.
- ▶ This is not an exhaustive list...many more factors such as directives of government, employment conditions, curiosity about new things, desire to understand causal relationships, social thinking and awakening, and the like may as well motivate (or at times compel) people to perform research operations.

Essential Skills for Research

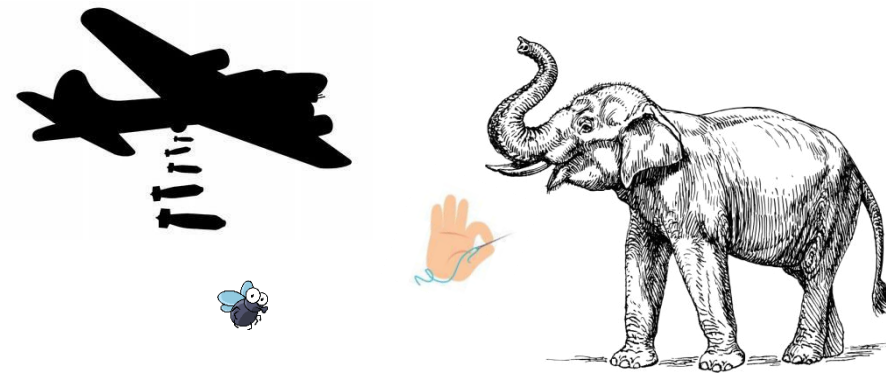
- ▶ In this 'Age of Information'
 - Information bombardment
- ▶ Essential Skills for Research includes
 - Searching Skills
 - How to find the 'right' information
 - Reviewing Skills
 - How to evaluate it
 - Engineering/Scientific Skills
 - How to develop/improve it
 - Writing Skills
 - How to report it clearly and accurately
 - Business Skills
 - How to make money out of it



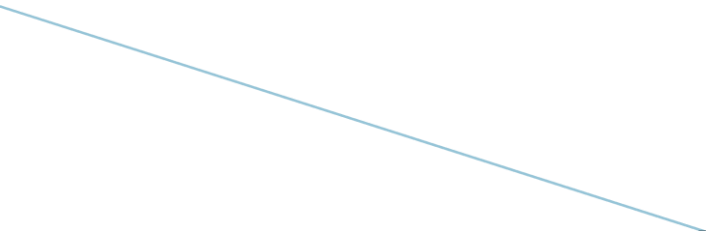
Today a person is subjected to more new information in a day than a person in the middle ages in his entire life!

Problem Solving Skills

- ✓ Right tool for the right problem
- ✗ When you have hammer in your hand then everything appear like a nail
- ✗ Dropping a bomb over a fly or hunting an elephant with a needle



What is NOT Research...

- ▶ Authority
 - ▶ Tradition
 - ▶ Common sense
 - ▶ Media myths
 - ▶ Personal experience
 - ▶ Rebuilding the existing system
 - ▶ Tweaks and tricks
 - ▶ ...
- 

Research Approaches

Research Methodologies

Research Approaches

- ▶ Research approaches are plans and the procedures for research that span the steps from broad assumptions to detailed methods of data collection, analysis, and interpretation.
- ▶ The selection of a research approach is based on many factors including the nature of the research problem
- ▶ Commonly used research approaches are
 - Quantitative
 - Qualitative
 - Mixed methods



Quantitative

Mixed methods

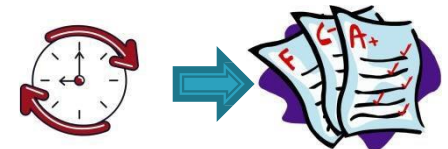
Qualitative

Quantitative

- ▶ Quantitative research is the process of collecting and analyzing numerical data.
- ▶ Objective study of the problem
 - Researcher keep himself/herself out of the problem situation
 - Bias controlled but have limited understanding of context
- ▶ Studies are often based on testing hypothesis
- ▶ Data is collected through experiments and surveys
- ▶ Statistical analysis is done for results validation
- ▶ The results can be generalized to broader populations based on the sampling method used.
- ▶ Quantitative research is widely used in the natural and social sciences e.g. biology, chemistry, physics, computer science, marketing etc.



Student
performance?



Study hours

Grades

Qualitative

- ▶ Qualitative research is the process of collecting and analyzing non-numerical data (e.g. text, audio or video).
- ▶ Subjective study of the problem
 - Researcher put himself/herself inside the problem situation in order to better understand the situation
 - Can hear the voices of individual but chances of biasness
- ▶ Data is often collected through interviews and observations
- ▶ Statistics can be used for analysis but not for results validation
- ▶ The results have limited generalizability.
- ▶ Qualitative research is commonly used in the humanities and social sciences e.g. sociology, education, health sciences, history etc.

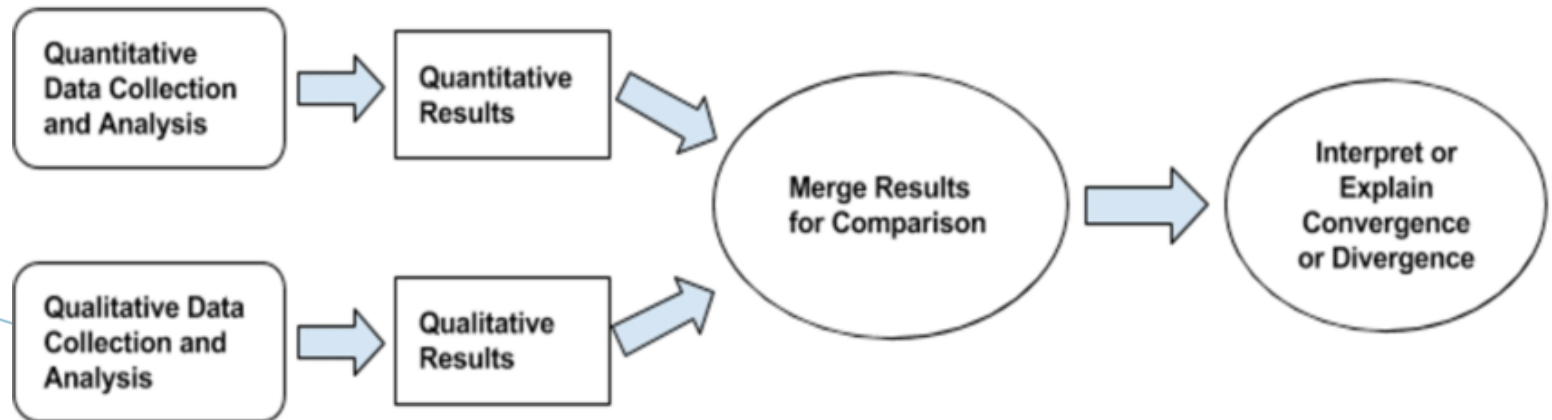


Student
performance?



Mixed methods

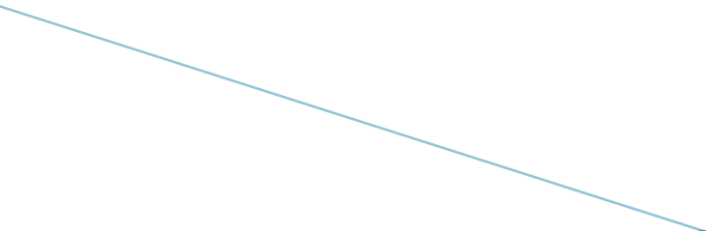
- ▶ Mixed method research is the process of collecting and analyzing both numerical and non-numerical data.
- ▶ Includes both objective and subjective study of the problem
- ▶ Combines both statistical analysis (quantitative) and personalized stories (qualitative)
- ▶ This method is used when the combination provides a better understanding of the problem
- ▶ Its not just about collecting and reporting quantitative and qualitative data separately, rather its about combining the two types of data



Research Designs

Research Methodologies

Research Designs

- ▶ Research designs are types of inquiry within qualitative, quantitative, and mixed methods approaches that provide specific direction for procedures
 - strategies of inquiry
 - ▶ Most commonly used research designs are
 - Quantitative design
 - Qualitative design
 - Mix Methods design
- 

Quantitative design

- ▶ Non-experimental designs, such as surveys, interviews
 - For example, we are interested in studying student behavior i.e. listening to music while studying.
- ▶ Experimental designs

These include

- true experiments and
- the less rigorous experiments called quasi-experiments

For example,

we are interested to know that whether or not listening to music have any effect on students performance?

◦ True Experiment

- Randomly divide student into two groups and assign them some task

e.g. memorization

- Group 1...doing this task while listening to music
- Group 2...doing this task without listening to music

◦ Quasi-experiments

when we don't have control on certain independent variables

- E.g. Effect of listening to music the performance of male and female students?

Qualitative design

▶ Narrative research

- Realizes heavily on the story telling (narratives) and personal experiences as a means of understanding the problem

▶ Phenomenology

- Researcher tries to describes the common shared lived experiences of individuals about a phenomenon (e.g. anger, professional growth etc.) as described by participants.

▶ Grounded theory

- This study design goes beyond description
- Researcher derives a general, abstract theory of a process, action, or interaction grounded in the views of participants.

Ethnographies

- Researcher studies the culture (values, beliefs, behaviors, language etc.) of a distinct group within the society.

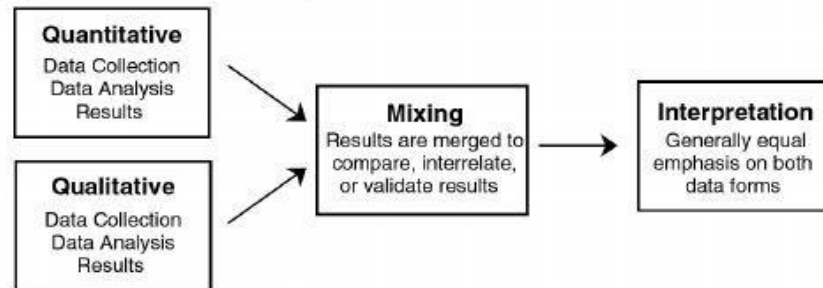
Case study

- Study of a particular real world case bounded in time/space
- Researcher develops an in-depth analysis of a case, often a program, event, activity, process, or one or more individuals

Mix Methods design

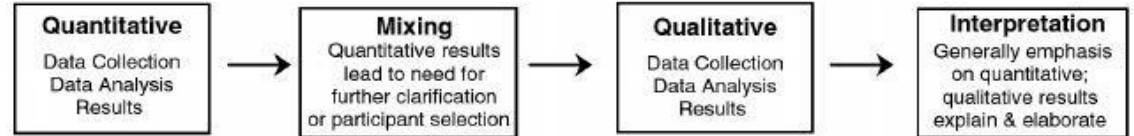
► Convergent

(a) Triangulation Design



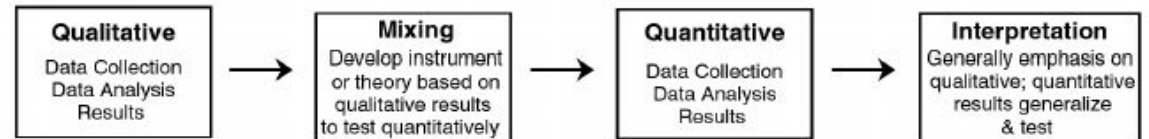
► Explanatory sequential

(b) Explanatory Design



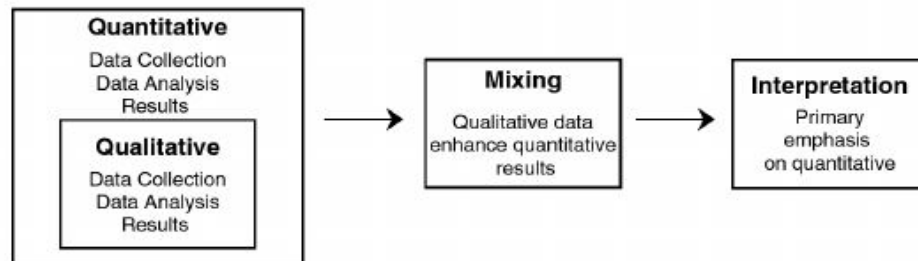
► Exploratory sequential

(c) Exploratory Design



► Transformative, embedded, or multiphase

(d) Embedded Design*



Research Methods

Research Methodologies

Research Methods

- ▶ The third major element in the research framework is the specific research methods that involve the forms of data collection, analysis, and interpretation
- ▶ Each of the research approach has its own associated research methods
 - Quantitative Methods
 - Qualitative Methods
 - Mixed Methods

Quantitative Methods



Data collection

- Pre-determined
- Instrument based questions (close-ended)
- Performance data, attitude data, observational data, and census data

▶ Analysis

- Statistical analysis

▶ Interpretation

- Statistical interpretation

Qualitative Methods

▶ Data collection

- Emerging methods
- Open-ended questions
- Interview data, observation data, document data, and audiovisual data

▶ Analysis

- Text and image analysis

▶ Interpretation

- Themes, patterns interpretation

Mixed Methods

▶ Data collection

- Both predetermined and emerging methods
- Both open- and closed-ended questions
- Multiple forms of data drawing on all possibilities

▶ Analysis

- Statistical and text analysis

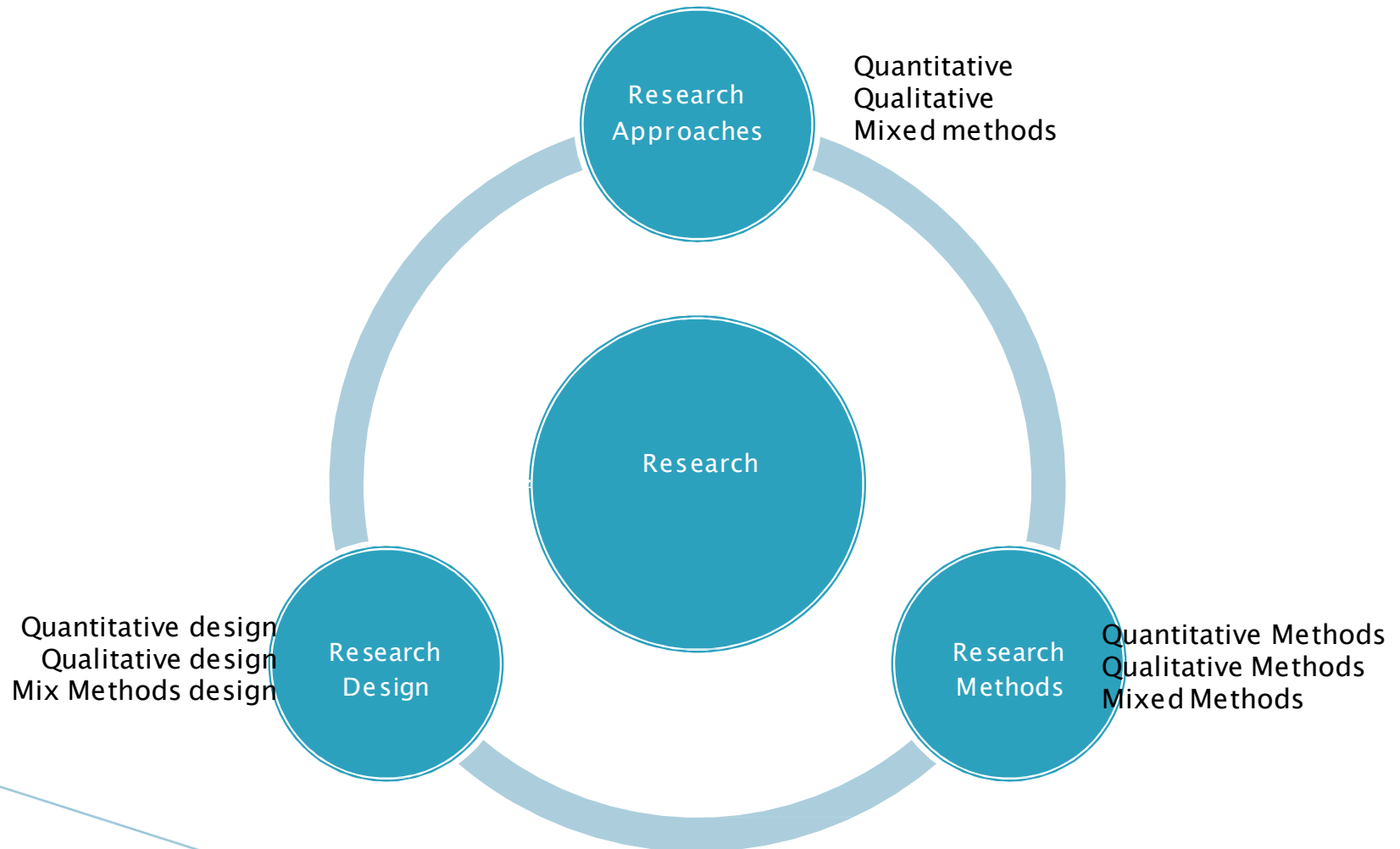
▶ Interpretation

- Across databases interpretation

Criteria for Selecting a Research Approach

Research Methodologies

The Interconnection of Worldviews, Design, and Research Methods



Criteria for Selecting a Research Approach

- ▶ Given the possibility of qualitative, quantitative, or mixed methods approaches, what factors affect a choice of one approach over another for the design of a proposal?
- ▶ Selection of a particular research approach depends upon
 - Researcher perspective/worldview,
 - Preferred research design,
 - Available research methods
 - Nature of research problem
 - The personal experiences of the researcher
 - Targeted audience(s)

Research in Computer Science

Research Methodologies

Research in Computer Science

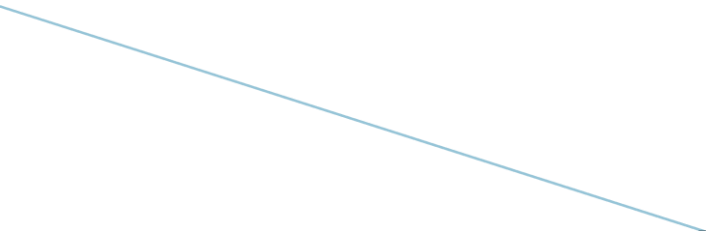
- ▶ Research in computer science can be carried out in different ways
 - Develop completely new idea/solution for a problem
 - e.g. Novel solutions
 - Suggest improvement in existing idea/solution
 - e.g. Enhanced solutions
 - Combine two or more ideas to develop a new idea
 - e.g. Hybrid solutions
 - Take idea/solution from one domain to solve a problem in another domain
 - e.g. Genetic algorithms

Research in Computer Science

► Methods to validate any new idea/solution

- Real experiments
- Simulations
- Mathematical/Analytical Modeling

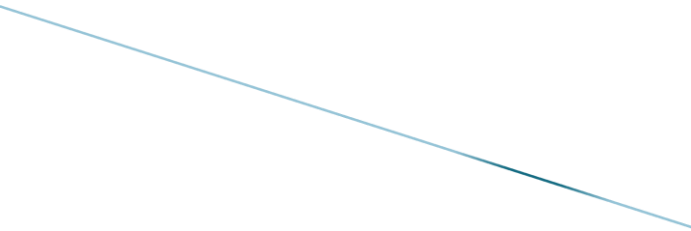
Real experiments

- ▶ Experiment are done on real data and devices.
 - ▶ This approach will give us the exact results
 - ▶ It involves too much labor and cost (in terms of time and money)
 - ▶ Slight modifications in the design but this approach is not flexible enough to accommodate these minor adjustments
 - ▶ This is not the best way to start with.
- 

Simulations

- ▶ This is done by performing simulations using simulators that closely reflect real world intended scenarios.
- ▶ Although, the simulator may not capture exactly all real world parameter and hence its results may be slightly different than true results but still it gives a nice insight of system behavior.
- ▶ The simulation models are easy to be developed and used to have a quick initial glance of system behavior with proposed modifications.
- ▶ Parameters can easily be fine-tuned to achieve optimal results at zero cost.
- ▶ Simulators can be used as an effective startup tool but its results cannot be fully trusted as they may not capture exact real world.

Mathematical / Analytical Modeling

- ▶ This is done by developing a mathematical model of the intended system and then the model can be used to evaluate the performance of a system under varying conditions to analyze its behavior.
 - ▶ These models are often based on certain assumptions regarding some of the system parameters which are often criticized and considered as flaw.
 - ▶ In reality, anything other than real implementation is based on some assumptions in one way or the other.
 - ▶ The assumptions are not just made blindly rather they are supported by strong and valid arguments.
 - ▶ Assumptions are based on closed approximation of the real world conditions.
- 

Summary

- ▶ Both simulation modeling and analytical modeling are considered as the most efficient way of doing initial performance analysis and often used in conjunction to validate each other.
 - ▶ They are useful where real system is not existing and yet to be developed.
 - ▶ Once a technique is proven working through modeling and simulation then its success is more likely in real implementations too.
- 